

∴ Maths:-

MC Qs:-

1. ~~A~~ B

2. B

3. A

4. B

5. ~~A~~ B

6. B

7. B

8. B

9. B

10. B

11. B

12. A

13. C

14. B

15. A

16. B

17. B

18. A

19. C

20. B

21. B

22. A

23. A

24. B

Q. 2 (i)

A : 2 (i)

(a) (i) $(\sqrt[4]{n^5})$

$$\frac{(n^5)^{\frac{1}{4}}}{n^{\frac{5}{4}}}$$

(ii) $(\sqrt[3]{a^2})$
 $(a^2)^{\frac{1}{3}}$
 $a^{\frac{2}{3}}$

(b) $\frac{n^3 \times n^5}{n^4}$

$$= \frac{n^8}{n^4}$$

$$= n^8 \div n^4$$

$$= n^4$$

Q. 2 (ii)

A : 2 (ii)

(a) $\log m + \log n - \log p$

$$\frac{\log mn}{p}$$

$$(b) \log_5 125$$

$$5 = 125$$

$$5 = (5)^3$$

$$n = 3$$

Q.2 (iii)

A.2 (iii)

$$7n - 4 < 24$$

$$7n < 24 + 4$$

$$7n < 28$$

$$n < \frac{28}{7}$$

$$n < 4$$

$$n \leq 4$$

$$\{n | n \in \mathbb{R}, n \leq 4\}$$

Q.3 (i)

$$(a) n^2 + 9n + 20$$

$$n^2 + 5n + 4n + 20$$

$$n(n+5) + 4(n+5)$$

$$(n+4)(n+5)$$

$$(b) 8a^3 + 27b^3$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$(2a)^3 + (3b)^3 = (2a+3b)(2a)^2 - (2a)(3b) + (3b)^2$$

$$(2a+3b)(4a^2 - 6ab + 9b^2)$$

Q HCF :-

$$8a^2b$$

$$12ab^3$$

$$4a^2b$$

$$4ab^3$$

$$\text{HCF} = 4ab$$

LCM :-

~~$$8 \times 3 = 24$$~~

$$8 \times 3 = 24$$

$$12 \times 2 = 24$$

taking bigger variables.
 $24a^2b^3$

Verify $\text{HCF} \times \text{LCM} =$

$$(4ab)(24a^2b^3)$$

$$96a^3b^4$$

$$\text{HCF} \times \text{LCM} = 8a^2b \times 12ab^3$$

$$96a^3b^4$$

HENCE PROVED.

Q: 3(iii)

A: 3(ii)

(a) $A(-1, 2)$ & $B(5, 10)$

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(5 - (-1))^2 + (10 - 2)^2}$$

$$= \sqrt{(5+1)^2 + (8)^2}$$

$$= \sqrt{(6)^2 + 64}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100}$$

$$= 10$$

(b) Midpoint of M

$$\frac{-1+5}{2}, \frac{2+10}{2}$$

$$\frac{4}{2}, \frac{12}{2}$$

$$M(2, 6)$$

(c) Distance AM :-

$$= \sqrt{(2 - (-1))^2 + (6 - 2)^2}$$

$$= \sqrt{(2+1)^2 + (4)^2}$$

$$= \sqrt{(3)^2 + 16}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25} \Rightarrow 5 \text{ units}$$

~~d BM~~ :-

$$\sqrt{(5-2)^2 + (10-6)^2} \left\{ \sqrt{9+16} \right.$$

$$= \sqrt{(3)^2 + (4)^2} \left\{ \sqrt{25} \Rightarrow 5 \right.$$